

NETWORKING & RECONNAISSANCE

Nmap Reconnaissance Scan Report & Analysis

DEMO / SAMPLE REPORT

NOT REAL SCAN RESULTS

All target details and scan outputs in this document are fictional examples created for academic demonstration.

Student Name	Sriram M
Department	Artificial Intelligence and Data Science
College	JNN Institution of Engineering
Task	Networking & Reconnaissance
Tools	Nmap, WHOIS, nslookup, Sublist3r
Target	192.0.2.10 (documentation-only sample IP)

Academic Demonstration • August 2026

1. Objective & Lab Environment

Objective: Illustrate the techniques of reconnaissance that allow the identification of hosts, ports, services, and potential indicators of attack surface within an authorized lab setting. Please note that this document is marked DEMO/SAMPLE. The IP addresses, domains, ports, services, and outputs listed are fictitious and should not be construed as any evidence of real system.

1.1 Scope

Item	Sample value
Target	192.0.2.10
Network	Documentation-only example network
Authorization	Assumed isolated academic lab
Purpose	Reconnaissance methodology demonstration

1.2 Tools Used

Tool	Purpose
WHOIS	Domain/IP registration information (where applicable)
nslookup	DNS record and name-resolution checks
Sublist3r	Subdomain enumeration demonstration
Nmap	Port, service, and OS reconnaissance

1.3 Ethical Boundary

Reconnaissance can only be performed on the systems where the person doing the testing holds proper authorization. For course purposes, the VM or CTF challenge provided by the instructors should be targeted. The target listed below is purely illustrative.

2. Passive Reconnaissance

Passive reconnaissance includes the collection of information without performing direct testing on the services on the target host. WHOIS and DNS records may provide some of the required information. The outputs listed below are fictitious.

2.1 WHOIS – Sample Observation

Command:

```
whois example-lab.test
```

Field	DEMO/SAMPLE result
Domain	example-lab.test
Registrar	Sample Registrar
Name Server	ns1.example-lab.test
Status	Example / fictional

2.2 nslookup – Sample Observation

Command:

```
nslookup example-lab.test
```

Sample output: Name: example-lab.test Address: 192.0.2.10

Interpretation: DNS resolution provides a hypothetical mapping between the sample hostname and documentation IP address. No claim is made about a real organization.

2.3 Sublist3r – Sample Observation

Command:

```
sublist3r -d example-lab.test
```

Sample subdomain	Purpose (illustrative)
www.example-lab.test	Web application
mail.example-lab.test	Mail service
dev.example-lab.test	Development environment

The listed subdomains are fictional examples intended to demonstrate how enumeration findings could be documented.

3. Nmap SYN Scan

SYN scans are typically conducted to detect open TCP ports by limiting the amount of connection establishment. Within the authorized lab, the tester can conduct Nmap scan to identify open TCP attack surface.

3.1 Command Used (Demo)

```
sudo nmap -sS 192.0.2.10
```

3.2 Fictional Scan Output

Port	State	Protocol	Sample observation
22	open	TCP	SSH
80	open	TCP	HTTP
443	open	TCP	HTTPS
3306	open	TCP	MySQL
8080	open	TCP	HTTP alternate

3.3 Analysis

The fictitious scan identifies five open TCP ports. SSH can be used to get administrative access. HTTP/HTTPS are web services, MySQL is possibly a database service and TCP/8080 is another web application. These are mere observations and do not by themselves constitute vulnerabilities.

3.4 Attack-Surface Summary

Category	Count	Priority for review
Web services	3	High
Remote administration	1	High
Database service	1	High

4. Service & Version Detection

Service/version detection attempts to determine what application is listening on an open port. In a real engagement, these results help defenders identify outdated software and prioritize patching.

4.1 Command Used (Demo)

```
nmap -sV 192.0.2.10
```

4.2 Fictional Results

Port	Service	Example version	Security review
22	OpenSSH	9.0	Check patch level and authentication policy
80	Apache HTTP Server	2.4.x	Review configuration and updates
443	Apache HTTPS	2.4.x	Check TLS configuration and certificates
3306	MySQL	8.0.x	Restrict network exposure
8080	HTTP application	Example/1.0	Review application security

4.3 Key Findings

- Multiple network-facing services increase the exposed attack surface.
- An exposed database port needs to be assessed and controlled whenever possible.
- Web services should be patched and securely configured.
- It would be inappropriate to use versioning for defensive asset inventory, but instead consider that it only indicates the vulnerability.

5. OS Detection & Network Topology

OS detection attempts to identify the operating-system family using network characteristics. Results can be uncertain and should be validated against authoritative asset information.

5.1 Command Used (Demo)

```
sudo nmap -O 192.0.2.10
```

5.2 Fictional OS Result

Item	DEMO/SAMPLE result
OS family	Linux (illustrative)
Confidence	Example only
Device role	Server / application host

5.3 Sample Network Topology

Tester / Kali Linux	→	Isolated Lab Network	→	Sample Target
10.10.10.5		10.10.10.0/24		192.0.2.10

Illustrative entry points: TCP/22 (SSH), TCP/80 (HTTP), TCP/443 (HTTPS), TCP/3306 (MySQL), and TCP/8080 (alternate HTTP).

Note: This topology is conceptual and does not represent a real network.

6. Security Analysis & Recommendations

The purpose of this section is to translate reconnaissance observations into defensive actions. Because the results are fictional, the recommendations are general security controls rather than confirmed vulnerability findings.

Observation	Potential concern	Recommended control
SSH exposed	Remote access surface	Use key-based authentication, restrict source networks, patch regularly
HTTP/HTTPS exposed	Web application attack surface	Patch server, secure headers/configuration, enforce HTTPS
MySQL exposed	Database exposure	Restrict access with firewall rules and bind to trusted interfaces
Port 8080 exposed	Additional application surface	Identify application owner and restrict unnecessary access
Version disclosure	Information leakage	Minimize unnecessary banners and maintain accurate patch inventory

6.1 Risk Prioritization

Priority	Meaning	Example action
High	Externally/unnecessarily exposed service	Restrict access and verify patching
Medium	Configuration or information exposure	Harden and monitor
Low	Minor information disclosure	Review during routine hardening

6.2 Important Limitation

Nmap reconnaissance alone does not establish that a host is exploitable. Vulnerability validation should be performed only with authorization and using an appropriate controlled testing process.

7. Conclusion & Evidence Summary

This DEMO/SAMPLE exercise demonstrates a structured reconnaissance workflow: passive information gathering, TCP port discovery, service/version identification, OS detection, topology mapping, and defensive analysis.

7.1 Consolidated Sample Findings

Area	Sample result
Target	192.0.2.10
Open TCP ports	22, 80, 443, 3306, 8080
Services	SSH, HTTP, HTTPS, MySQL, alternate HTTP
OS estimate	Linux – illustrative
Primary review areas	Remote access, web services, database exposure

7.2 Evidence Checklist

- WHOIS / passive reconnaissance screenshot
- nslookup screenshot
- Sublist3r screenshot
- Nmap SYN scan screenshot
- Nmap service/version scan screenshot
- Nmap OS detection screenshot
- Network topology diagram
- Security analysis and recommendations

7.3 Final Statement

This document is an academic DEMO/SAMPLE. The scan results and the network structure presented herein are purely fictional. They should be replaced with actual data obtained by you during an authorized lab exercise.

Prepared by: Sriram M
Artificial Intelligence and Data Science
JNN Institution of Engineering